# TWO DIMENSION FUNCTIONS ON A LOCALLY NOETHERIAN SOBER SPACE DIFFER BY A LOCALLY CONSTANT FUNCTION

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-topology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-topology/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

*Editorial note.* Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

*Editorial note.* Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

*Allowed standard prerequisites (editorial).*

- The statement that a Noetherian space has finitely many irreducible components. Invoked inside the counted proof (source lines 4124-4126). Provides the finitely many components through the point.
- The additive formula for the values of a dimension function along the codimension of specializations. Invoked inside the counted proof (source lines 4132-4136). Rewrites the difference of the two functions on a component.

*Source reference.* Original lemma lemma-Noetherian: sources/stacks-topology/pinned-original-source.tex, source lines 1470–1470. Navigation pointer only: the counted text refers to the Stacks item lemma-Noetherian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original lemma lemma-dimension-function-catenary: sources/stacks-topology/pinned-original-source.tex, source lines 4083–4083. Navigation pointer only: the counted text refers to the Stacks item lemma-dimension-function-catenary, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

**Lemma 0.1.** *Let $X$ be a topological space. Let $\delta$, $\delta'$ be two dimension functions on $X$. If $X$ is locally Noetherian and sober then $\delta - \delta'$ is locally constant on $X$.*

*Date*: stacks-project:topology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* Let $x \in X$ be a point. We will show that $\delta - \delta'$ is constant in a neighbourhood of $x$. We may replace $X$ by an open neighbourhood of $x$ in $X$ which is Noetherian. Hence we may assume $X$ is Noetherian and sober. Let $Z_1, \ldots, Z_r$ be the irreducible components of $X$ passing through $x$. (There are finitely many as $X$ is Noetherian, see Lemma lemma-Noetherian.) Let $\xi_i \in Z_i$ be the generic point. Note $Z_1 \cup \ldots \cup Z_r$ is a neighbourhood of $x$ in $X$ (not necessarily closed). We claim that $\delta - \delta'$ is constant on $Z_1 \cup \ldots \cup Z_r$. Namely, if $y \in Z_i$, then

$$\delta(x) - \delta(y) = \delta(x) - \delta(\xi_i) + \delta(\xi_i) - \delta(y) = -\text{codim}(\overline{\{x\}}, Z_i) + \text{codim}(\overline{\{y\}}, Z_i)$$

by Lemma lemma-dimension-function-catenary. Similarly for $\delta'$. Whence the result. $\square$